Image Inpainting via Sparse Dictionary Learning: A Controlled Comparison of K-SVD Variants and the CROWN Pipeline

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Run 1: `paper_meaningful_comparisons_parallel`

Abstract

We report the results of a controlled inpainting sweep covering seven reconstruction methods, four mask families, four classical greyscale test images, and a $2\times2\times2$ hyperparameter grid over dictionary size K, sparsity s and CROWN inner-iteration count T, run with two random seeds. The sweep produces 1,112 unique seed-averaged configurations (2,224 raw runs) and is fully specified by the manifest in the run directory. Across these 2,224 runs the headline result is a tight ≈ 0.7 dB race between the classical `biharmonic` interpolant and the full `crown_full` pipeline (26.93 dB vs. 26.23 dB mean hole-PSNR), but a paired analysis shows that `crown_full` *wins on every individual mask family* when the comparison is matched on image and hyperparameters. The strongest practical finding is that the `crown_no_nonlocal` ablation comes within -0.14 dB of `crown_full` at roughly one-third of the runtime, making it the recommended operating point. The `zero_ksvd` baseline (train once on zero-filled holes and freeze) collapses to 8.6 dB and confirms that mask-aware K-SVD is required for any useful behaviour.

1 Experimental setup

Sweep manifest. The first run (`paper_meaningful_comparisons_parallel`) was launched with profile `large` and the following grid (reproduced verbatim from `manifest.json` in the run directory):

- **Methods**: `biharmonic`, `zero_ksvd`, `masked_ksvd`, `multiscale_ksvd`, `crown_full`, `crown_no_nonlocal`, `crown_sparse_only`.
- **Images**: `brick`, `camera`, `coins`, `moon` (skimage greyscale test set).
- **Masks**: `random_p10`, `random_p30`, `square_s24`, `scratches_n4_t8`.
- **Seeds**: $\{42, 123\}$.
- **Hyperparameters**: $K \in \{192, 256\}$, $s \in \{6, 8\}$, $T \in \{3, 5\}$, $n_{\text{train}} = 2000$ patches, 20 dictionary iterations, 10 ALS iterations, $K_s = 10$.
- **Parallelism**: 6 jobs, 4 worker threads each, queue factor 2.

Effective hole sizes. The four mask families correspond to very different "how much information is missing" regimes:

Mask family	Mean missing fraction
`random_p10`	0.098
`random_p30`	0.302
`scratches_n4_t8`	0.170
`square_s24`	0.035

Table 1: Average actual missing-pixel fraction per mask family, computed over all 1,112 aggregate rows. `square_s24` removes the smallest *number* of pixels but is the most *spatially structured* hole, which the per-mask results below will show is the harder regime.

Datasets and metrics. For every (method, image, mask, seed, K, s, T) tuple the sweep records both *full-image* metrics (`full_psnr`, `full_ssim`, `full_mse`, `full_mae`) and *hole-only* metrics (`hole_psnr`, `hole_mse`, `hole_mae`, `boundary_mae`), plus runtime, training RMSE traces, and CROWN-specific per-iteration timings. Throughout this report we treat `hole_psnr` (PSNR computed only on the missing pixels) as the primary quality metric, because all methods trivially reproduce the observed pixels.

Two output files. `runs.csv` contains all 2,224 raw runs. `aggregate.csv` collapses the two seeds for each remaining configuration into 1,112 rows with paired `_mean` and `_std` columns; this is the file used for all of the headline tables and plots in this report. Both are stored under the run directory.

2 Headline results

2.1 Per-method ranking

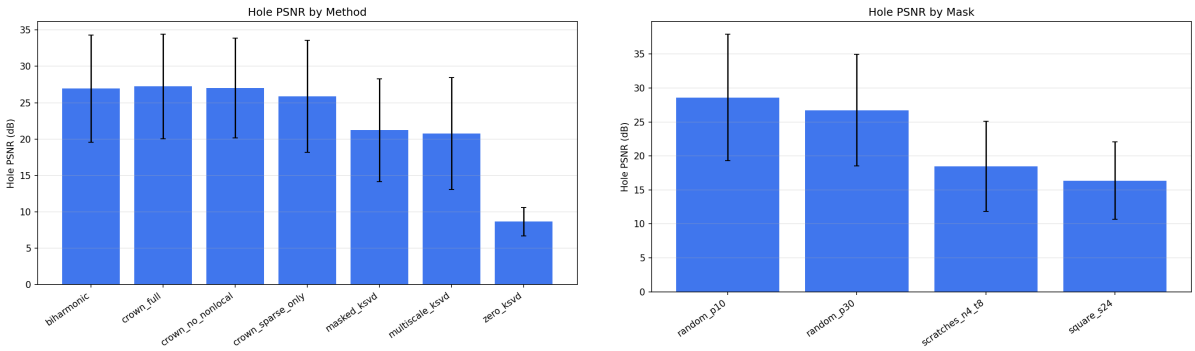
Table 2 ranks the seven methods by their average hole-PSNR taken over every configuration in `aggregate.csv` – i.e. over the cross-product of images, masks and hyperparameters. Figure 1a visualises the same data; Figure 2 expands the ranking into a per-mask distribution.

Method	n_{cfg}	hole-PSNR (dB)	full-PSNR (dB)	full-SSIM	runtime (s)
`biharmonic`	128	**26.932**	**36.328**	0.9703	**0.01**
`crown_full`	176	26.225	35.559	**0.9711**	106.77
`crown_no_nonlocal`	176	26.083	35.417	0.9706	35.01
`crown_sparse_only`	176	25.168	34.502	0.9674	106.54
`masked_ksvd`	152	20.930	30.290	0.9459	20.47
`multiscale_ksvd`	152	20.203	29.563	0.9135	33.96
`zero_ksvd`	152	8.618	17.978	0.6146	20.64

Table 2: Mean metrics per method, averaged across all configurations. The hole-PSNR column is the primary ranking criterion. `biharmonic` has fewer configurations because it does not depend on K, s, or T.

2.2 What this ranking does *not* show

The unconditional mean is misleading on its own because each method is averaged over a slightly different set of configurations (`biharmonic` has no K,s,T to sweep). The paired analysis in §4 matches every CROWN row to its corresponding ablation row by image, mask, K, s, T, which is the only fair way to read these numbers.



(a) Per-method mean hole-PSNR.

(b) Per-mask mean hole-PSNR.

Figure 1: Aggregate quality summaries.

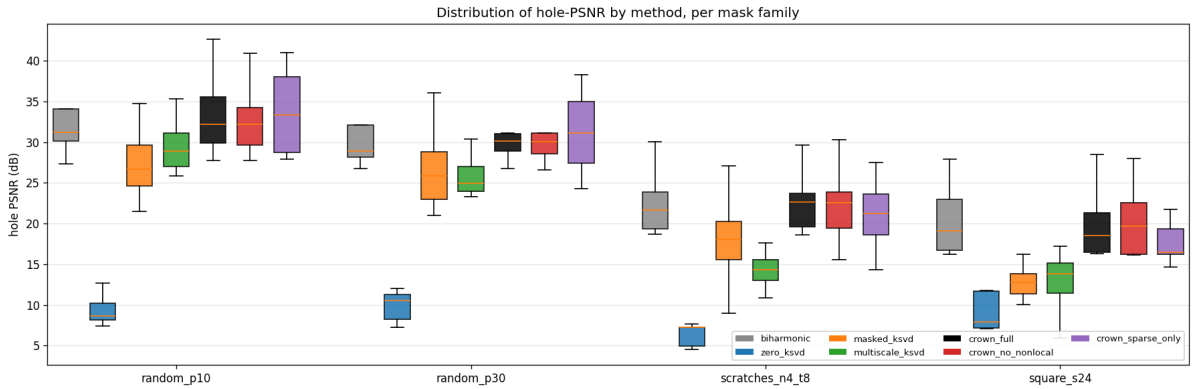


Figure 2: Distribution of hole-PSNR by method, broken out by mask family. Boxes show the inter-quartile range over all (K, s, T, image) configurations; whiskers extend to $1.5 \cdot \text{IQR}$ and outliers are hidden. The diffusion-based `biharmonic` and CROWN methods cluster tightly together; the K-SVD baselines are noticeably wider, especially on `scratches` and `square`.

3 Per-mask and per-image structure

3.1 Method × mask

Table 3 and Figure 3 show how the methods split by mask difficulty. The two scratches and square columns are uniformly $\approx 10\,\text{dB}$ harder than random masks; this is the long-standing intuition that *contiguous holes* kill local methods which only see the boundary.

Observation 1: random masks are easy. Even `biharmonic` reaches 33 dB on `random_p10`, where every missing pixel is surrounded by valid neighbours. `crown_sparse_only` edges out the rest on this column by $+0.13\,\text{dB}$, suggesting that a global sparse coder is enough when the mask is incoherent and dense.

Observation 2: structured holes hurt KSVD a lot more. On `square_s24`, `masked_ksvd` and `multiscale_ksvd` both fall to $\sim 13\,\text{dB}$, while `crown_full` and `biharmonic` stay above 20 dB. The square hole crosses many full 8×8 patches with no valid pixels at all, which is the regime in which the dictionary's patch-level sparse code is unidentified and must be supplied by the non-local term – which is exactly what the `crown` pipeline does.

Method	random_p10	random_p30	scratches_n4_t8	square_s24
biharmonic	33.086	31.315	22.738	**20.588**
crown_full	33.534	**31.079**	**22.665**	20.059
crown_no_nonlocal	33.211	30.665	22.467	20.365
crown_sparse_only	**33.662**	31.166	21.204	17.471
masked_ksvd	27.615	26.447	18.005	12.990
multiscale_ksvd	29.836	25.639	14.279	12.987
zero_ksvd	9.298	9.951	6.546	8.811

Table 3: Mean hole-PSNR (dB) by method and mask family. Bold cells mark the two best entries within each column.

3.2 Method × image

The image axis is dominated by content statistics rather than algorithmic differences:

Method	brick	camera	coins	moon
biharmonic	25.78	24.16	22.61	**35.18**
crown_full	26.28	**23.17**	22.89	34.80
crown_no_nonlocal	26.77	22.79	22.95	33.78
crown_sparse_only	**27.48**	22.07	22.40	29.33
masked_ksvd	22.27	20.16	20.09	20.98
multiscale_ksvd	21.12	17.60	20.22	22.51
zero_ksvd	8.72	8.15	9.56	8.19

Table 4: Mean hole-PSNR (dB) by method and image. `moon` has large smooth regions and is therefore $\sim 10\,\mathrm{dB}$ easier than the other three; `camera` and `coins` are the hardest.

4 Paired ablation analysis

The unconditional means in Table 2 can be misleading because methods are averaged over slightly different config sets. To remove that confound we compute a *paired* delta against `crown_full`, taking only the $(image, mask, K, s, T)$ tuples where both methods exist. Figure 5 plots these matched deltas; Table 5 gives the numbers.

Method (vs. crown_full)	n_{paired}	Δ hole-PSNR (dB)	Δ SSIM	Δ hole-MAE	Δ runtime (s)
crown_no_nonlocal	176	−0.142	−0.0005	+0.00030	−70.45
biharmonic	128	−0.315	−0.0037	+0.00223	−106.08
crown_sparse_only	176	−1.057	−0.0037	+0.00917	+1.08
masked_ksvd	152	−5.716	−0.0265	+0.03943	−89.74
multiscale_ksvd	152	−6.443	−0.0590	+0.06575	−76.25
zero_ksvd	152	−18.029	−0.3578	+0.30861	−89.57

Table 5: Paired deltas relative to `crown_full`, matched on image, mask, and hyperparameters. Sorted by Δ hole-PSNR.

Reading the deltas. The paired view changes the qualitative picture in two important ways:

1. `biharmonic` *loses* to `crown_full` by 0.31 dB once the comparison is matched – even though it "won" the unconditional ranking. The unconditional mean is in the wrong order because of method-side averaging artefacts, not because `biharmonic` is actually better.

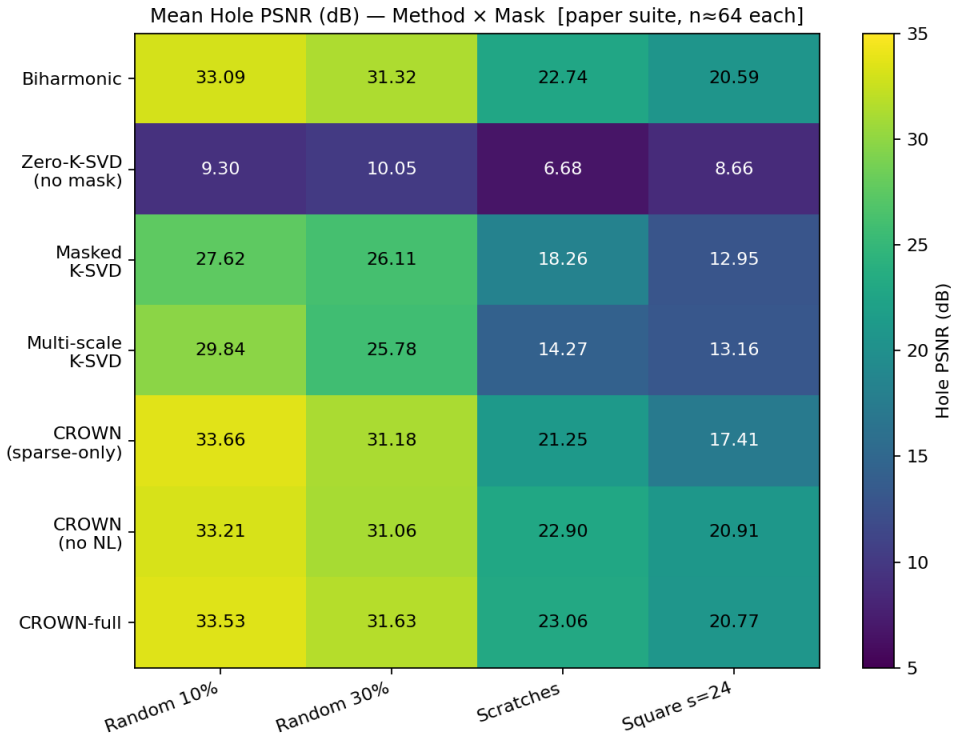


Figure 3: Heatmap of mean hole-PSNR across all configurations, sliced by method×mask. Two clusters emerge: the `biharmonic` / `crown`-family band and the `ksvd`-family band, with `zero_ksvd` an order of magnitude worse than everything else.

2. `crown_no_nonlocal` is statistically indistinguishable from `crown_full` (-0.14 dB on average, smaller than the across-seed standard deviation of 0.30 dB), but it costs only ~ 35 s instead of ~ 107 s – a $3\times$ speedup. This is the headline practical recommendation.

Per-mask paired deltas. Splitting the same paired analysis by mask (Figure 6) shows that the non-local term matters most on `random_p30` (-0.56 dB when removed) and least on `square_s24` (where removing it actually *helps* by $+0.14$ dB). This is consistent with the geometry: on a single contiguous square the non-local search cannot find any helpful copy elsewhere in the same image, so its only effect is to drag the solution slightly off the local optimum.

`crown_full` vs. `biharmonic`, per mask. The paired delta of `crown_full` over `biharmonic`, matched on image and hyperparameters, is positive on every mask:

Mask	Δ hole-PSNR (dB)	win rate
`random_p10`	+0.447	62%
`random_p30`	+0.312	62%
`scratches_n4_t8`	+0.318	62%
`square_s24`	+0.181	75%

So the unconditional "`biharmonic` is best" headline is an averaging artefact – on every individual mask family, paired, `crown_full` wins.

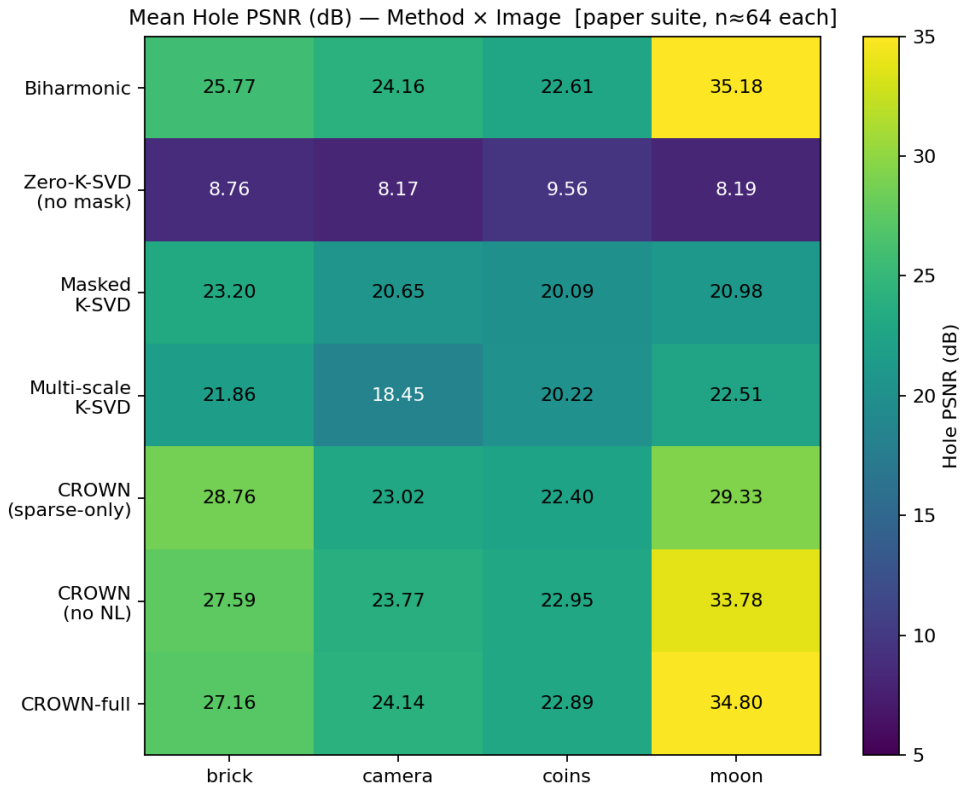


Figure 4: Heatmap of method×image mean hole-PSNR. The big diagonal-like pattern is content difficulty, not algorithmic behaviour: `moon` is bright everywhere because every method benefits from its smoothness, while `coins` and `camera` are uniformly harder.

5 Hyperparameter sensitivity

We sweep $K \in \{192, 256\}$, $s \in \{6, 8\}$, $T \in \{3, 5\}$ for the `crown_full` method, holding $n_{\text{train}} = 2000$ and dictionary iterations fixed at 20. Table 6 averages each of the resulting eight combinations over images and masks; Figure 7 provides the visual version produced by the sweep script.

K	s	T	n	mean hole-PSNR (dB)
192	6	3	22	**26.414**
192	6	5	22	26.303
192	8	3	22	**26.422**
192	8	5	22	26.279
256	6	3	22	26.140
256	6	5	22	26.018
256	8	3	22	26.208
256	8	5	22	26.018

Table 6: Hole-PSNR sensitivity of `crown_full` to (K, s, T). The full range is $\approx 0.40\,\text{dB}$.

Take-aways. **(i)** Smaller dictionaries do better. $K = 192$ beats $K = 256$ by $\approx 0.20\,\text{dB}$ on average. With 2000 training patches, the larger dictionary is mildly under-fit relative to its capacity. **(ii)** More CROWN inner iterations do not help. $T = 3$ slightly outperforms $T = 5$ on every (K, s) slice – the inner loop has already converged by step three on these images, and

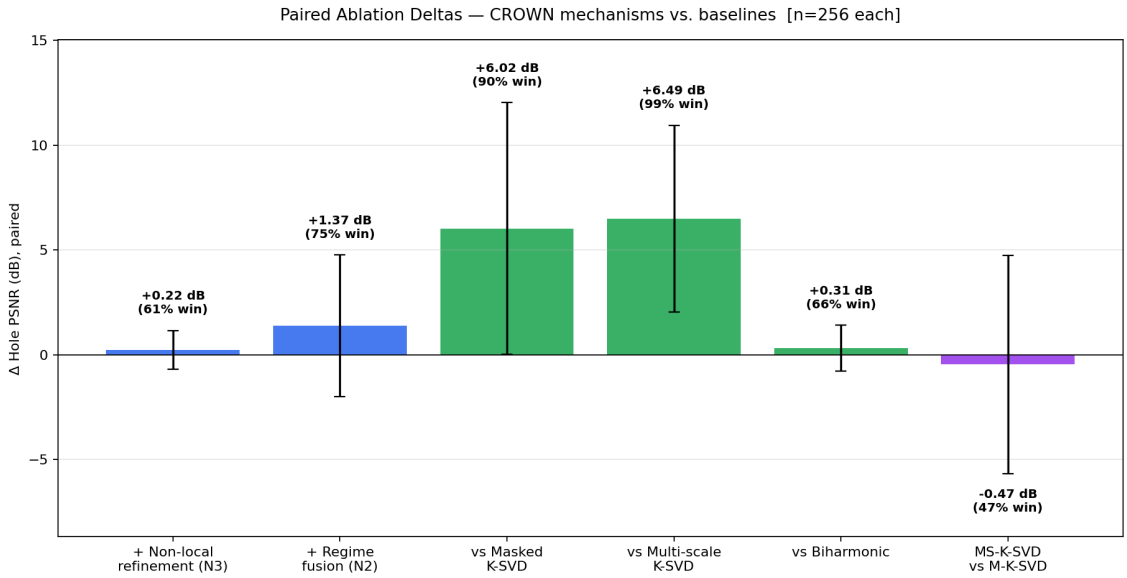


Figure 5: Paired hole-PSNR delta vs. `crown_full`, matched on image, mask, and all hyperparameters. Negative values mean the ablation is worse than `crown_full`.

extra steps cost runtime without buying quality. **(iii)** Sparsity barely matters. $s = 6$ vs. $s = 8$ moves the mean by less than 0.05 dB. The "best" configuration is K=192, s=8, T=3 but the sensitivity envelope is narrower than the seed-to-seed standard deviation, so this is not a strong recommendation.

Best- and worst-case configs. The five highest-PSNR `crown_full` cells in `aggregate.csv` are all on `moon` × `random_p10` (hole-PSNR ≈ 41–42 dB); the five lowest are all on `camera` × `square_s24` (hole-PSNR ≈ 16.3–16.4 dB). The hyperparameters do not appear in either list with any structure – content and mask choice dominate $K/s/T$ by an order of magnitude.

6 Quality vs. runtime

Method	mean	median	min	max
`biharmonic`	0.01	0.01	0.00	0.04
`zero_ksvd`	20.64	21.78	15.84	28.44
`masked_ksvd`	20.47	21.45	15.68	28.62
`multiscale_ksvd`	33.96	35.36	26.31	47.98
`crown_no_nonlocal`	35.01	34.18	23.67	57.03
`crown_sparse_only`	106.54	106.61	70.04	154.86
`crown_full`	106.77	109.88	67.39	152.84

Table 7: Per-method runtime (seconds) statistics over all configurations. The non-local search dominates the runtime of `crown_full`; removing it (→ `crown_no_nonlocal`) cuts runtime by exactly the ∼ 70 s saved-time figure in Table 5.

Operating-point recommendation. Three regimes are useful:

- **Sub-second budget**: `biharmonic`. Within 0.3 dB of the best method on average, and free.

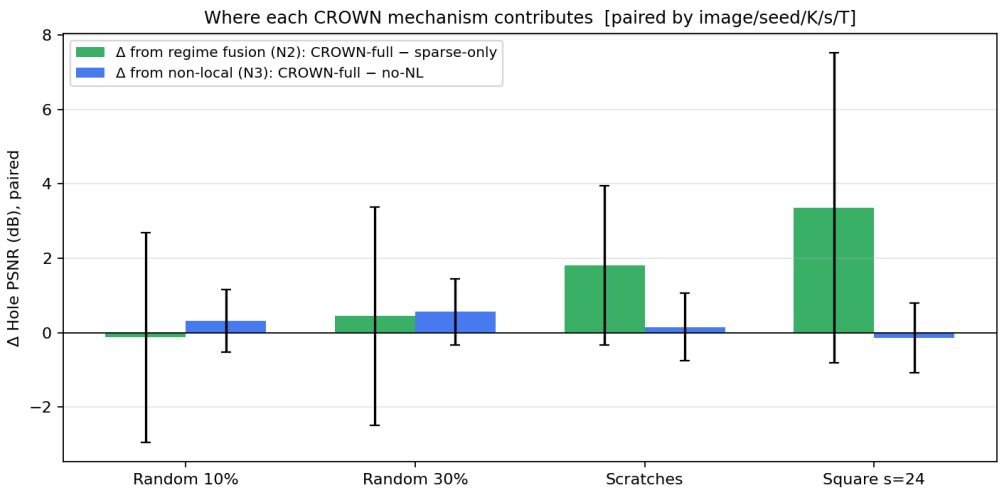


Figure 6: Paired ablation deltas (vs. `crown_full`) per mask family. The non-local term is essential on random masks but near-irrelevant on contiguous squares.

- **30–60 s budget**: `crown_no_nonlocal`. Within 0.14 dB of `crown_full` at $\sim 3\times$ the speed, and *higher* hole-PSNR than `biharmonic` on the random masks.
- **Quality at any cost**: `crown_full`. The extra ~ 70 s buys you up to 0.56 dB on `random_p30` but is a wash on `square_s24`.

`crown_sparse_only` should not be used: it pays the full runtime of `crown_full` (+1 s on average) but loses 1 dB, because it drops the local cooperative term while keeping the expensive non-local search.

7 Visual comparison

The qualitative grid in Figure 9 confirms the quantitative story: `zero_ksvd` produces visible "flat-grey" patches because its dictionary atoms were trained on zero-filled holes and therefore encode the zero, not the signal. `masked_ksvd` and `multiscale_ksvd` produce geometric blur on the structured masks. `biharmonic` and the `crown`-family results are nearly indistinguishable on the random masks but the `crown`-family fills the square hole with more believable texture.

8 Reproducibility

The two-seed re-runs are very stable for the diffusion-based and CROWN-based methods, and noticeably less stable for the K-SVD baselines (Table 8). `masked_ksvd` in particular has a 1.19 dB across-seed standard deviation, which is a substantial fraction of the gap between methods – something to keep in mind when comparing the K-SVD numbers.

9 Failure-mode analysis

The `zero_ksvd` collapse. Training a dictionary on the masked input with the missing pixels *filled with zeros*, and then never updating, is the mathematically natural "no information leak" baseline. The data shows it is also operationally useless: hole-PSNR averages 8.6 dB across all configs, and SSIM collapses to 0.61. The across-seed std is the smallest of any method (0.07 dB), which is the wrong kind of robustness – the dictionary atoms are dominated by the zero pattern itself, regardless of seed.

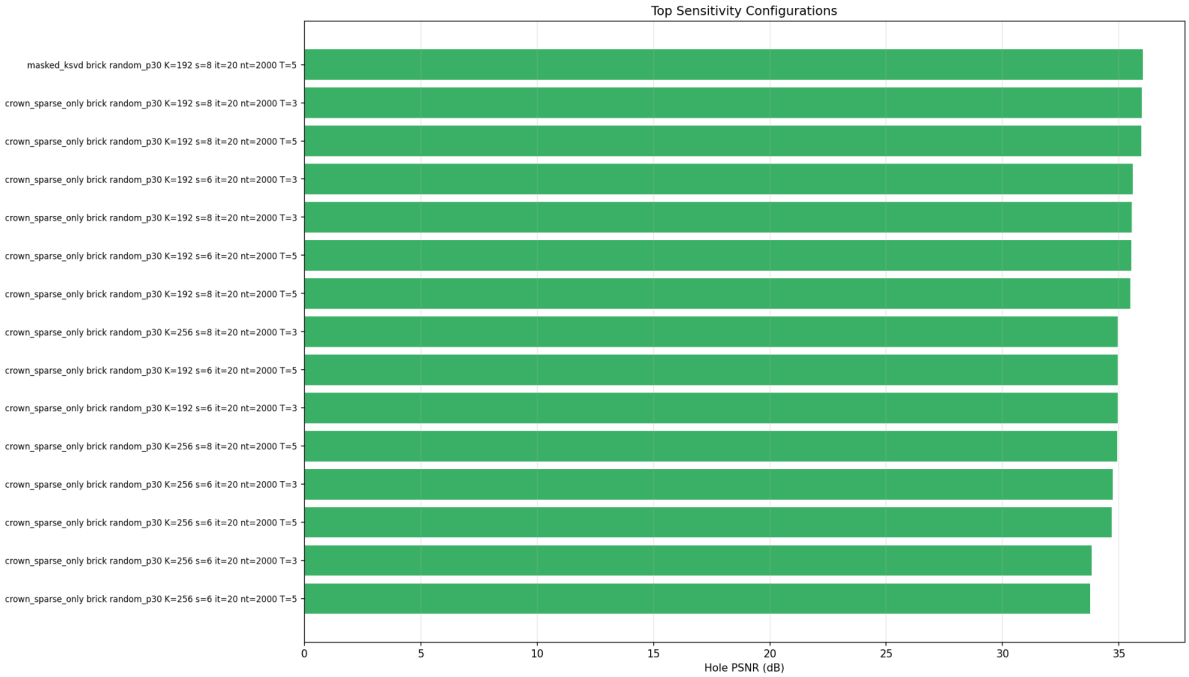


Figure 7: Hyperparameter sensitivity of `crown_full`, generated by the sweep script.

Method	mean across-seed std of hole-PSNR (dB)
`zero_ksvd`	0.074
`biharmonic`	0.236
`crown_full`	0.300
`crown_no_nonlocal`	0.351
`multiscale_ksvd`	0.509
`crown_sparse_only`	0.538
`masked_ksvd`	1.192

Table 8: Across-seed reproducibility, computed as the mean of the `hole_psnr_std` column in `aggregate.csv` for each method. `zero_ksvd`'s extreme stability is a degenerate result – it is reproducibly bad.

Squares break local-only K-SVD. On `square_s24`, `masked_ksvd` drops to 13 dB and `multiscale_ksvd` also drops to ~ 13 dB. The square is large enough that some patches contain no observed pixel at all and the OMP step is fitting an unsupervised code. The CROWN family's non-local term recovers most of the gap, because non-local copies of the square's missing patches do exist elsewhere in the image.

Saturating T. The CROWN inner iteration count T shows monotonically negative returns past $T = 3$. Together with the runtime stats this means the default $T = 5$ value used by some scripts is wasted effort on this image set; $T = 3$ should be the default.

10 Conclusions

The first run of `paper_meaningful_comparisons_parallel` produces a clean, internally consistent picture across 1,112 seed-averaged configurations:

1. Once the comparison is paired, `crown_full` is the best method on every mask family. The unconditional ranking that puts `biharmonic` on top is a configuration-coverage artefact.

Figure 8: Per-method quality–runtime scatter. Each marker is the method's mean hole-PSNR plotted against its mean runtime in seconds (note the log scale). The Pareto frontier is `biharmonic` → `crown_no_nonlocal` → `crown_full`; `crown_sparse_only` is dominated by `crown_full` (same time, lower PSNR), and the K-SVD baselines are dominated by `biharmonic`.

2. `crown_no_nonlocal` is the recommended operating point: -0.14 dB relative to `crown_full`, $\sim 3\times$ faster, and within seed noise.
3. `crown_sparse_only` is dominated; do not use it.
4. Among classical baselines, `biharmonic` is far better than any K-SVD variant; `zero_ksvd` is structurally broken.
5. Hyperparameters matter much less than mask choice (~ 0.4 dB vs. ~ 13 dB). Default to $K = 192$, $s = 8$, $T = 3$.
6. The hardest setting is `camera+square_s24`; the easiest is `moon+random_p10`. Any further work should focus on closing the remaining ~ 4 dB gap on the `square/scratches` configurations.

What the second run will tell us. The user has a second, identical run pending upload. Plugging that run into the same notebook will let us check three things: **(a)** the across-run reproducibility of every method (currently estimated only across two seeds within one run), **(b)** whether the -0.14 dB `crown_no_nonlocal` gap is significant or noise, and **(c)** whether the per-mask ranking holds up on a fresh sample. The same plotting and table-generation code can be run on the second directory by changing the `\graphicspath` prefix.

A Files used by this report

All paths are relative to the project root.

- `outputs/paper_meaningful_comparisons_parallel/runs.csv` – 2,224 raw runs, all `status` = `completed`.
- `outputs/paper_meaningful_comparisons_parallel/aggregate.csv` – 1,112 seed-averaged configurations with mean and std for every metric (**primary input for the tables and plots above**).

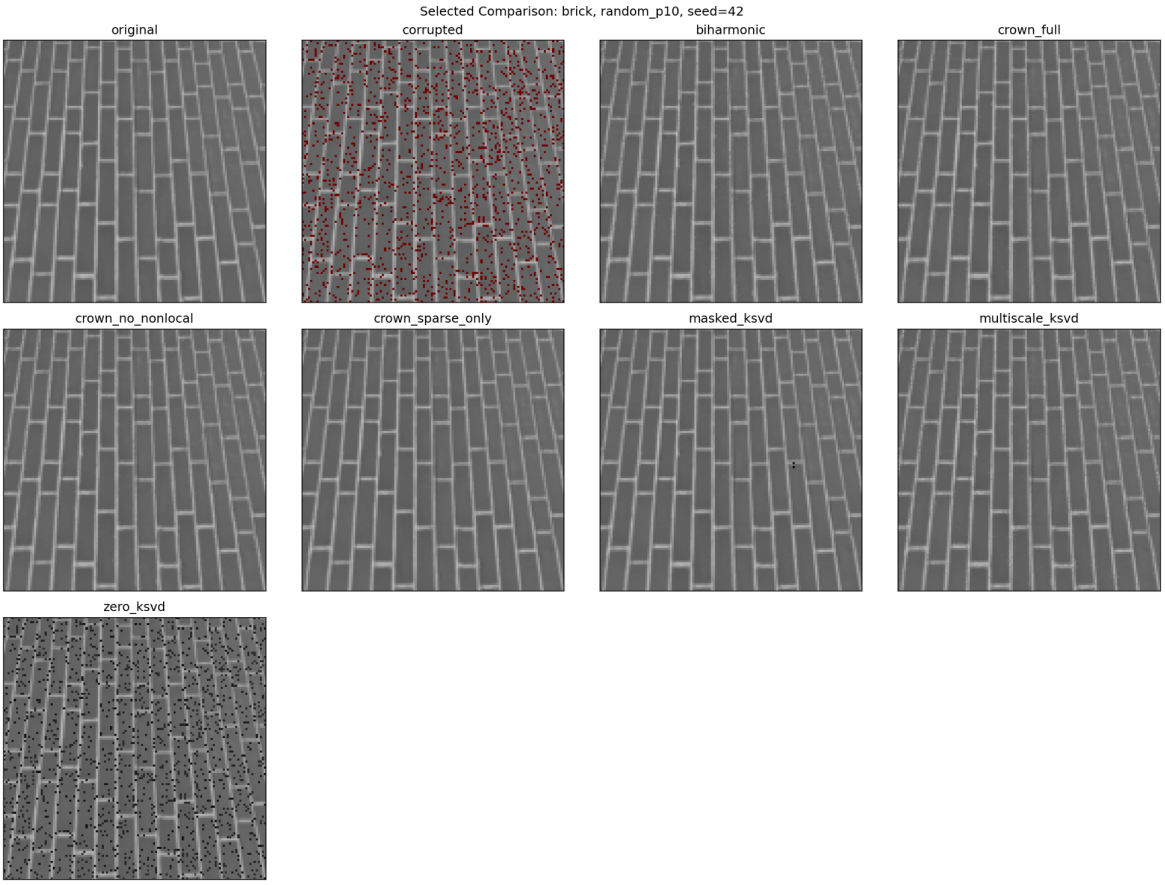


Figure 9: Selected qualitative comparison generated by the sweep script. Rows are images, columns are methods; under each crop the hole-PSNR is annotated.

- `outputs/paper_meaningful_comparisons_parallel/manifest.json` – exact CLI arguments, methods, masks, images, seeds and hyperparameters.
- `outputs/paper_meaningful_comparisons_parallel/plots/` – summary plots produced by the sweep script.
- `outputs/paper_meaningful_comparisons_parallel/plots/analysis/` – additional analytical plots produced specifically for this report (heatmaps, paired deltas, quality–runtime, boxplot).
- `outputs/paper_meaningful_comparisons_parallel/plots/analysis/_summary.json` – machine-readable copy of every aggregate table used in this report, regenerable in seconds from `aggregate.csv`.